

I. Criteria for choosing the right enterprise for the project?

- Clear Bounded Scope (The "Town Planner" View):

- Choose the enterprise that is complex enough to have a recognizable division of labor and multiple IT systems.
- Ideal size: A mid-sized regional business (50–1,000 employees), such as a wholesale distributor (e.g., *ABC Trading*), a regional hospital/healthcare clinic (e.g., *Academic Diagnostic Medicine*), or a logistics provider.
- What to avoid: Global conglomerates (too vast, and complex) or single-owner retail shops (no real IT footprint, making EA useless).

- An Explicit Business Catalyst (The "Why"):

- The chosen enterprise must face a pressing business driver (e.g., a strategic shift to online sales, migrating legacy infrastructure to the cloud, or integrating systems after a merger).
- "Speak business, not tech" and ensures every technical architectural decision you make can be traced back to business strategy.

- Understandable Business and IT Landscapes:
 - Easy to identify at least 2–4 core business capabilities (like Customer Management or Logistics) and the applications supporting them. If you cannot find or reasonably mock up this baseline, you will struggle with the subsequent ADM phases.

II. Deliverables for a Two-Student Team

Work balance assignment can be split into specialized roles:

- Student A (The Organizational Architect)
- Student B (The Framework & Metamodel Architect)
- Student A and B - Core Principles.

Deliverable 1: The Organizational Model for EA (Student A Lead)

Focuses on scoping the architecture capability, mapping stakeholders, and aligning with existing business frameworks.

- **Boundary and Scope Statement:** Define exactly which departments, organizational units, and geographic locations are impacted by the EA team.
- **Stakeholder Map and RACI (Responsibility Assignment) Matrix:** Identify the key stakeholders (CEO, CFO, CIO, and business unit heads), detail their key concerns, and

establish a RACI matrix defining who is Responsible, Accountable, Consulted, and Informed for the ADM activities.

- **Framework Interface Assessment:** Describe how the EA team will interact with other existing management frameworks in the business (e.g., matching the ADM cycle with their Agile/Scrum sprints or ITIL service operations).

Deliverable 2: The Tailored Architecture Framework (Student B Lead)

Focuses on customizing the TOGAF metamodel and content framework to fit the specific enterprise.

- **Tailored Enterprise Metamodel:** Review the standard TOGAF Foundation Metamodel and customize it for the company by selecting relevant entities and deleting irrelevant ones.
- **Tailored Content Framework:** Define which specific catalogs, matrices, and diagrams from the Extended Content Framework the team will commit to producing during the subsequent ADM phases. For example, you might mandate a *Business Capability Map* for Phase B and an *Application Communication Diagram* for Phase C.

Deliverable 3: Collaborative Foundations (Group)

- **The Principles Catalog (3–5 Core Principles):** Draft a small set of foundational Enterprise and Architecture Principles. Each principle must follow the strict TOGAF template: **Name, Statement, Rationale, and Implications**.
- **Request for Architecture Work:** A formal, high-level briefing document from the business sponsors triggering the ADM cycle, summarizing the business goals, strategic drivers, and major organizational constraints.

Guide on Creating a Principle Catalog

1. Enterprise Principles vs. Architecture Principles

- **Enterprise (or Business) Principles:** High-level guidelines that shape the overall business strategy, corporate culture, and organizational decision-making (e.g., customer focus, business agility, or sustainability).

- **Architecture Principles:** More specific, technology-oriented guidelines that focus on how the IT landscape and systems are designed and integrated to support those business goals (e.g., reusability, platform independence, or standardization).

2. The Standard TOGAF 4-Part Template

1. **Name:** A short, memorable, and unambiguous title (it should not mention specific product or vendor names).
2. **Statement:** A concise, clear, and unambiguous declaration of the fundamental rule.
3. **Rationale:** The business justification for why this principle is important and what benefits it brings, using business terminology.
4. **Implications:** The practical impact on both the business and IT (in terms of costs, resource allocation, and tasks). This answers the critical question: *"How does this affect me?"*

3. Examples of Principles

Example A: "Common Vocabulary and Definitions" (Data Domain)

- **Name:** Common Vocabulary and Definitions
- **Statement:** Data must be defined consistently throughout the organization, and those definitions must be understandable and available to all users.
- **Rationale:** To build integrated systems and share information, the data used in the development of applications must have a single, agreed-upon definition across the business.
- **Implications:**
 - The business must dedicate resources to establish and maintain a unified corporate glossary of descriptions.
 - Whenever a new system is built or a new data element is introduced, the team must coordinate and reconcile its definitions with the master glossary.

Example B: "Comply with Laws and Regulations" (Business/Governance Domain)

- **Name:** Comply with Laws and Regulations
- **Statement:** All information management and architectural processes must comply with all relevant laws, external policies, and regulations.

- **Rationale:** Non-compliance exposes the organization to severe legal liabilities, financial penalties, and loss of public trust.
- **Implications:**
 - The organization must stay strictly aligned with data protection, privacy, and regulatory rules (such as HIPAA, GDPR, etc.).
 - Technical convenience, speed, and cost savings can never override legal requirements; changes in external laws may force direct modifications to processes or applications.

4. How to Evaluate a "Good" Set of Principles

When grading your students' submissions, you can evaluate their Principles Catalog against these **five TOGAF quality criteria**:

- **Completeness:** Do the principles cover all major situations the enterprise will face?
- **Robustness:** Are they definitive enough to guide tough, controversial design decisions?
- **Understandability:** Can a non-technical business stakeholder quickly grasp their intent?

- **Consistency:** Are they free of direct contradictions (where following one would force you to violate another)?
- **Stability:** Are they enduring enough to survive the semester without needing weekly rewrites?

⇒ If your students are working on a specific case study, I can help you draft a custom set of 3–5 tailored architecture principles that perfectly fit their chosen business scenario.

In the TOGAF® standard, **Architecture Principles** are the fundamental, general rules and guidelines that govern and direct all architecture work and decision-making within an organization. They are initially developed by Enterprise Architects in collaboration with key stakeholders and formally ratified by the **Architecture Board** during the **Preliminary Phase**. Because they act as the "constitution" of the architecture practice, they are designed to be stable, enduring, and strategic—meaning they change very rarely and carry a strong binding character for all future development projects.

1. Enterprise Principles vs. Architecture Principles

It is important to help your students distinguish between these two levels of principles:

- **Enterprise (or Business) Principles:** High-level guidelines that shape the overall business strategy, corporate culture, and organizational decision-making (e.g., customer focus, business agility, or sustainability).
- **Architecture Principles:** More specific, technology-oriented guidelines that focus on how the IT landscape and systems are designed and integrated to support those business goals (e.g., reusability, platform independence, or standardization).

2. The Standard TOGAF 4-Part Template

Every architecture principle your students write must be formalized using a strict **four-part structure**. This prevents them from being vague, "wishlist" items and forces students to think about the real-world impact of their architectural choices:

1. **Name:** A short, memorable, and unambiguous title (it should not mention specific product or vendor names).
2. **Statement:** A concise, clear, and unambiguous declaration of the fundamental rule.
3. **Rationale:** The business justification for why this principle is important and what benefits it brings, using business terminology.

4. **Implications:** The practical impact on both the business and IT (in terms of costs, resource allocation, and tasks). This answers the critical question: *"How does this affect me?"*
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3. Concrete Examples of Principles

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3. Data is an Asset (Data Domain)

- **Name:** Data is an Asset
- **Statement:** Data is a valuable corporate asset that has real, measurable worth and must be managed accordingly.

- **Rationale:** Accurate, complete, and timely data is the foundation of effective decision-making. Treating data as a corporate asset ensures that the organization actively invests in its quality, protection, and accessibility.
- **Implications:**
 - The business must assign clear data stewardship and domain-driven ownership to prevent data silos.
 - Data must be actively profiled, normalized, and validated at the ingest boundaries to maintain its value over time.
 - Sufficient budget, physical resources, and security controls must be allocated to lifecycle data management.

4. Technology Independence (Application/Technology Domain)

- **Name:** Technology Independence
- **Statement:** Software applications must be independent of specific technology platforms and be able to operate across diverse environments.
- **Rationale:** Tying applications tightly to a specific operating system, proprietary database, or hardware vendor makes the enterprise vulnerable to obsolescence and

vendor lock-in. Ensuring independence maximizes software longevity and operational flexibility.

- **Implications:**

- Development teams must build interfaces using standard web APIs, microservices, and open protocols rather than vendor-specific extensions.
- Applications and data structures must be decoupled from the underlying physical infrastructure.
- If a technology provider discontinues support or modifies pricing, the system must be modular enough to be migrated with minimal disruption.

5. Control Technical Diversity (Technology/Governance Domain)

- **Name:** Control Technical Diversity
- **Statement:** The variety of technologies, development languages, and hardware platforms supported across the enterprise must be strictly limited.
- **Rationale:** Supporting multiple different technologies for similar processing environments introduces non-trivial maintenance costs, complicated integrations, and a massive training burden for IT staff. Limiting diversity simplifies maintainability and reduces the Total Cost of Ownership (TCO).

- **Implications:**

- The organization must establish and enforce a clear Standards Library and Technology Standards Profile.
- Before introducing a new database, cloud service, or framework, teams must conduct objective due diligence to prove that existing corporate standards cannot satisfy the business use case.
- Training programs can be standardized, and technical asset reuse can be maximized across the organization.

6. Business Continuity (Business/Governance Domain)

- **Name:** Business Continuity
- **Statement:** Enterprise operations and critical customer-facing services must not be disrupted by technical failures or system outages.
- **Rationale:** Modern enterprises rely heavily on continuous digital availability to maintain daily operations. Unexpected system interruptions lead to immediate financial losses, compliance penalties, and a severe loss of customer trust and brand loyalty.
- **Implications:**

- High availability (HA), disaster recovery (DR), and backup-archiving procedures must be designed into the physical and logical architectures from the start.
- Critical hardware and databases must be structured with redundant components and automated failover capabilities.
- The organization must conduct regular "fire drills" and simulate system failures to proactively verify that SLAs can be met under stress.

7. Maximize Benefit to the Enterprise (Business Domain)

- **Name:** Maximize Benefit to the Enterprise
- **Statement:** All architectural decisions and technical investments must be guided by what brings the greatest return on investment and strategic value to the entire organization, rather than to a single department.
- **Rationale:** Organizations are naturally divided into business silos, which can lead to redundant software purchases and misaligned IT spending. Decisions made at the enterprise level ensure synergy, cost-efficiency, and long-term strategic alignment.
- **Implications:**

- Individual departments may have to concede their local preferences or custom software requirements for the greater benefit of the company.
- Standardization and shared services (such as a unified customer portal) are heavily prioritized over building bespoke point solutions.
- Every IT initiative must be justified through a corporate business case that explicitly maps the work back to the enterprise's strategic goals.

Common diagrams/artifacts you produce in TOGAF ADM

[TOGAF Diagrams – Free TOGAF 10 Enterprise Architecture & ADM Diagram Library](https://nelsambrose.github.io/togaf-diagrams/)
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Phase A: Architecture Vision

- **Stakeholder map / stakeholder diagram** (who needs what from the architecture)
- **Architecture vision diagram** (high-level representation of the intended outcomes)
- **Value chain / value stream style view** (often light, to frame “why” and “what”)

Phase B: Business Architecture

- **Business capability map / capability diagram**
- **Business process diagrams** (BPMN-style or equivalent)
- **Organization / role structure diagram** (org units, roles, responsibilities)
- **Information/interaction views for business** (how major business components interact)

Phase C: Information Systems Architecture

Phase C1 (Data)

- **Data entity diagram / logical data model** (entities, attributes, relationships)
- **Data services or data domain diagrams** (logical groupings and ownership)

- **Data flow / data mapping view** (where data is created/used)

Phase C2 (Applications)

- **Application landscape diagram** (apps and their relationships)
- **Application interaction diagram** (integration/communication between apps)
- **Application-to-capability mapping** (apps realizing business capabilities)

Phase D: Technology Architecture

- **Technology architecture diagram / platform view**
- **Technology capabilities map** (platform components and responsibilities)
- **Deployment and hosting view** (where software runs—often at least a logical version)
- **Integration/communications topology diagram** (networks, middleware, major interfaces)

Phase E: Opportunities & Solutions

- **Target-state architecture view(s)** (the “what it will look like” visuals)
- **Solution option diagrams** (alternative solution structures and how they differ)
- **Roadmap visualization** (often diagrammatic: timeline, increments, epics)
- **Capability/implementation grouping diagram** (what goes together as major increments)

Phase F: Migration Planning

- **Migration / transition diagram**
- **Implementation sequencing diagram** (dependencies and order)
- **Gap-to-solution mapping view** (baseline → target and what changes)

Phase G: Implementation Governance

- **Architecture compliance / governance diagrams** (decision workflow and checkpoints)
- **Requirements traceability views** (requirements mapped to architecture deliverables)
- **Deviation handling view** (how exceptions are raised/approved)

Phase H: Architecture Change Management

- **Change impact diagrams** (what aspects are affected by the change)
- **Updated baselines/targets view diagrams** (reflect what changed after feedback)

Views

- **Business view diagrams** (capabilities, processes, org)

- **Data view diagrams** (logical data model, domains)
- **Application view diagrams** (application landscape and interactions)
- **Technology view diagrams** (platform and deployment topology)
- **Transition view diagrams** (roadmap + migration sequencing)

What's “required” vs “optional”?

TOGAF gives guidance on deliverables/contents; organizations typically tailor diagrams. A practical way to treat “needed” is:

- **Required for understanding/approval:** diagrams that show baseline vs target and the main relationships (capabilities → processes, apps → capabilities, data entities → applications, tech platforms → apps/integration, plus migration sequencing).
- **Optional for depth:** detailed BPMN, very granular deployment diagrams, etc., only when your audience needs that granularity.

Core ADM deliverables (baseline)

- **Architecture Building Blocks** (produced in F, H; consumed in A, B, C, D, E)
- **Architecture Contract**
- **Architecture Definition Document** (produced in B, C, D, E, F; consumed in C, D, E, F, G, H)
- **Architecture Principles** (produced and consumed across Preliminary, A, B, C, D, ... depending on tailoring)
- **Architecture Repository** (created/used from Preliminary through the ADM execution)
- **Architecture Requirements Specification** (produced in B–F; used in Requirements Management and later phases)
- **Architecture Roadmap** (produced in B–F; used within B–F)
- **Architecture Vision** (produced in A and E; used in later phases)
- **Business Principles, Business Goals, and Business Drivers** (produced in Preliminary/A/B; consumed in A/B)
- **Capability Assessment** (produced in A and E; used in B–F)
- **Change Request** (produced in F, G, H)
- **Communications Plan** (produced in A; used in B–F)
- **Compliance Assessment** (produced in G; consumed in H)
- **Implementation and Migration Plan** (produced in E, F; consumed in F)

- **Implementation Governance Model** (produced in F; consumed in G, H)
- **Organizational Model for Enterprise Architecture** (produced in Preliminary; used in A–H)
- **Request for Architecture Work** (produced in Preliminary/F/H; used in A and G)
- **Requirements Impact Assessment**
- **Solution Building Blocks** (produced in G; used in A–G)
- **Statement of Architecture Work** (produced across A–H; used as inputs for Requirements Management and later consumption)
- **Tailored Architecture Framework**

Roadmap to ADM Deliverables

Source: Andrew Josey, Dave Hornford - The TOGAF Standard, 10th Edition. A Pocket Guide (2022, Van Haren Publishing)

Page171/220

Table 54. Roadmap to ADM Deliverables

ADM Phase	Reference(s)
Preliminary Phase	Section 7.1 – Tailored Architecture Framework Section 7.2 – Organizational Model for Enterp Architecture Section 7.3 – Architecture Principles Section 7.4 – Business Principles, Business Go and Business Drivers Section 7.5 – Request for Architecture Work
Phase A: Architecture Vision	Section 7.6 – Statement of Architecture Work Section 7.7 – Architecture Vision Section 7.8 – Communications Plan Section 7.9 – Capability Assessment Section 7.10 – Architecture Definition Document

Phase B: Business Architecture	Section 7.10 – Architecture Definition Document Section 7.11 – Architecture Requirements Specification Section 7.12 – Architecture Roadmap Section 7.13 – Architecture Building Blocks
Phase C: Information Systems Architectures	Section 7.10 – Architecture Definition Document Section 7.11 – Architecture Requirements Specification Section 7.12 – Architecture Roadmap Section 7.13 – Architecture Building Blocks
Phase D: Technology	Architecture Section 7.10 – Architecture Definition Document Section 7.11 – Architecture Requirements Specification Section 7.12 – Architecture Roadmap Section 7.13 – Architecture Building Blocks
Phase E: Opportunities and Solutions	Section 7.10 – Architecture Definition Document Section 7.13 – Architecture Building Blocks Section 7.12 – Architecture Roadmap Section 7.14 – Solution Building Blocks Section 7.15 – Implementation and Migration Plan Section 7.16 – Transition Architecture Section 7.17 – Implementation Governance Model
Phase F: Migration Planning	Section 7.12 – Architecture Roadmap Section 7.15 – Implementation and Migration Plan Section 7.16 – Transition Architecture Section 7.17 – Implementation Governance Model

Phase G: Implementation Governance	Section 7.17 – Implementation Governance Model Section 7.18 – Architecture Contracts Section 7.19 – Change Request Section 7.20 – Compliance Assessment
Phase H: Architecture Change Management	Section 7.17 – Implementation Governance Model Section 7.18 – Architecture Contracts Section 7.19 – Change Request Section 7.20 – Compliance Assessment Section 7.5 – Request for Architecture Work Section 7.21 – Requirements Impact Assessment
ADM Architecture Requirements Management	Section 7.11 – Architecture Requirements Specification Section 7.21 – Requirements Impact Assessment

ADM Phase Purpose

P53/174

ADM Phase	Purpose
Preliminary Phase	Describes the preparation and initiation activities required to create an Architecture Capability, including customization of the TOGAF framework and definition of Architecture Principles.
Requirements Management	Operates the process of managing architecture requirements throughout the ADM. This is a continuous phase which ensures that any changes to requirements are handled through appropriate

	governance processes and reflected in all other phases.
Phase A: Architecture Vision	Describes the initial phase of an architecture development cycle. It includes information about defining the scope of the architecture development initiative, identifying the stakeholders, creating the Architecture Vision, and obtaining approval to proceed with the architecture development.
Phase B: Business Architecture Phase C: Information Systems Architectures (Data and Application) Phase D: Technology Architecture	Describes the development of four architectures, that are commonly accepted as subsets of an overall Enterprise Architecture, to support the agreed Architecture Vision: — Business — Information Systems – Data — Information Systems – Application — Technology
Phase E: Opportunities and Solutions	Conducts initial implementation planning and the identification of delivery vehicles for the architecture defined in the previous phases.
Phase F: Migration Planning	Addresses how to move from the Baseline to the Target Architectures by finalizing a detailed Implementation and Migration Plan.

Phase G: Implementation Governance	Provides architectural oversight for the implementation.
Phase H: Architecture Change Management	Establishes procedures for managing change to the new architecture.